

BEC305 Python Programming — Syllabus mapping

EC branch, semester 3, 2025 scheme. Prescribed textbook: Al Sweigart, *Automate the Boring Stuff with Python*, No Starch Press, 1st edition, 2015.

Course outcomes

CO	Statement
CO1	Acquire programming skills in Python.
CO2	Demonstrate data structure representation using Python.
CO3	Develop the skill of pattern matching and files in Python.
CO4	Acquire Object Oriented Skills in Python.
CO5	Develop the ability to write database applications in Python.

Module plan

Module	Title	Hours	Chapters	Units	Deck slides	COs
1	Python Basics, Flow Control and Functions	9	TB1 (Sweigart 1e) Ch. 1 Python Basics, Ch. 2 Flow Control, Ch. 3 Functions, pp. 13-78	16	56	CO1
2	Data Structures: Lists, Dictionaries and Strings	8	TB1 (Sweigart 1e) Ch. 4 Lists, Ch. 5 Dictionaries and Structuring Data, Ch. 6 Manipulating Strings, pp. 79-144	16	56	CO2
3	Pattern Matching with Regular Expressions and File Handling	8	TB1 (Sweigart 1e) Ch. 7 Pattern Matching with Regular Expressions, Ch. 8 Reading and Writing Files, pp. 147-192	16	56	CO3
4	Classes and Objects	9	TB2 (Downey, Think Python 2e) Ch. 15-18 — TB2 was not supplied in this environment, so chapters are cited as the syllabus gives them and no page numbers are claimed	16	56	CO4
5	Networked Programs, Web Services and Databases	8	TB3 (Severance, Python for Everybody) Ch. 12, Ch. 13 up to 13.5, Ch. 15 — TB3 was not supplied in this environment, so chapters are cited as the syllabus gives them and no page numbers are claimed	16	56	CO5

Module 1 — Python Basics, Flow Control and Functions

Driving question: How do I turn an idea into a program that makes decisions, repeats work, and can be broken into pieces I can reuse?

Topic to chapter, pages and deck slides

#	Teaching unit	Chapter	Pages	Deck slides	CO	Bloom
1	What Python is and what the interactive shell is for	Ch. 1	13-16	5, 6, 7	CO1	L2
2	The integer, floating-point and string data types	Ch. 1	16-17	8, 9, 10	CO1	L2
3	String concatenation and replication	Ch. 1	17-18	11, 12, 13	CO1	L3
4	Storing values in variables: assignment statements and names	Ch. 1	18-21	14, 15, 16	CO1	L2
5	Your first program, dissected	Ch. 1	21-23	17, 18, 19	CO1	L2
6	The print(), input(), len(), str(), int() and float() functions	Ch. 1	23-28	20, 21, 22	CO1	L3
7	Boolean values and comparison operators	Ch. 2	32-35	23, 24, 25	CO1	L2
8	Boolean operators and mixing them with comparisons	Ch. 2	35-37	26, 27, 28	CO1	L3
9	Elements of flow control: conditions, blocks and program execution	Ch. 2	37-38	29, 30, 31	CO1	L2
10	if, else and elif statements	Ch. 2	38-45	32, 33, 34	CO1	L3
11	while loops, break and continue	Ch. 2	45-53	35, 36, 37	CO1	L3
12	for loops and the range() function	Ch. 2	53-57	38, 39, 40	CO1	L3
13	Importing modules and ending a program early with sys.exit()	Ch. 2	57-60	41, 42, 43	CO1	L3
14	def statements, parameters, return values and None	Ch. 3	61-65	44, 45, 46	CO1	L3
15	Keyword arguments, local and global scope, and the global statement	Ch. 3	65-72	47, 48, 49	CO1	L4
16	Exception handling and the Guess the Number program	Ch. 3	72-78	50, 51, 52	CO1	L4

Verbatim syllabus lines

- Python Basics, Python language features, History, Entering Expressions into the Interactive Shell
- The Integer, Floating-Point, and String Data Types, String Concatenation and Replication
- Storing Values in Variables, Your First Program, Dissecting Your Program
- Flow control, Boolean Values, Comparison Operators, Boolean Operators, Mixing Boolean and Comparison Operators
- Elements of Flow Control, Program Execution, Flow Control Statements
- Importing Modules, Ending a Program Early with sys.exit()
- Functions, def Statements with Parameters, Return Values and return Statements, The None Value
- Keyword Arguments and print(), Local and Global Scope, The global Statement
- Exception Handling, A Short Program: Guess the Number
- Text book 1: Chapters 1-3
- Number of Hours: 9

Provenance

Built from TB1 Chapters 1-3, which the syllabus names for this module. The organising idea is that a program is expressions plus three control structures plus functions, and that almost every beginner bug is a type error, an indentation error or a scope error rather than a logic error. The interactive shell is used throughout as the place to test one expression before committing it to a file. Exception handling is introduced here rather than later because the Guess the Number program needs it.

Module 2 — Data Structures: Lists, Dictionaries and Strings

Driving question: Which of the four built-in containers should hold my data, and what does choosing wrongly actually cost me?

Topic to chapter, pages and deck slides

#	Teaching unit	Chapter	Pages	Deck slides	CO	Bloom
1	The list data type and indexing	Ch. 4	80-82	5, 6, 7	CO2	L2
2	Negative indexes and slicing	Ch. 4	82-83	8, 9, 10	CO2	L3
3	Changing lists: len(), assignment by index, concatenation and del	Ch. 4	83-84	11, 12, 13	CO2	L3
4	Working with lists: for loops, in and not in	Ch. 4	84-87	14, 15, 16	CO2	L3
5	Multiple assignment and augmented assignment	Ch. 4	87-89	17, 18, 19	CO2	L4
6	List methods: index, append, insert, remove and sort	Ch. 4	89-92	20, 21, 22	CO2	L3
7	Mutable and immutable types, and the tuple	Ch. 4	93-97	23, 24, 25	CO2	L3
8	References, copy() and deepcopy()	Ch. 4	97-102	26, 27, 28	CO2	L4
9	The dictionary data type, and dictionaries versus lists	Ch. 5	105-109	29, 30, 31	CO2	L3
10	get(), setdefault() and checking for keys	Ch. 5	109-111	32, 33, 34	CO2	L4
11	Pretty printing and modelling real-world things	Ch. 5	111-113	35, 36, 37	CO2	L4
12	Nested dictionaries and lists	Ch. 5	113-119	38, 39, 40	CO2	L4
13	Working with strings: literals, indexing and slicing	Ch. 6	123-127	41, 42, 43	CO2	L3
14	Case methods and the isX methods	Ch. 6	127-131	44, 45, 46	CO2	L3
15	startswith, endswith, join and split	Ch. 6	131-133	47, 48, 49	CO2	L3
16	Justifying and stripping text, and the clipboard with pyperclip	Ch. 6	133-142	50, 51, 52	CO2	L4

Verbatim syllabus lines

- Data Structures: Lists: The List Data Type, Working with Lists
- Strings: Manipulating Strings, Working with Strings, Useful String Methods
- Tuples and Dictionaries, basics
- Using Data Structures to Model Real-World Things
- Manipulating Strings
- Text book 1: Chapters 4-6
- Number of Hours: 8

Provenance

Built from TB1 Chapters 4-6, which the syllabus names for this module. The spine is the mutable-versus-immutable distinction, introduced early and then used to explain every later surprise: why a list passed to a function can be changed by it, why a tuple cannot, why `copy.deepcopy` exists, and why every string method returns a new string rather than editing in place. Dictionaries are presented as the answer to "look it up by meaning rather than position", with `get()` and `setdefault()` as the two methods that remove most `KeyError` handling.

Module 3 — Pattern Matching with Regular Expressions and File Handling

Driving question: How do I describe a shape of text precisely enough for a program to find it, and then get that text safely into and out of a file?

Topic to chapter, pages and deck slides

#	Teaching unit	Chapter	Pages	Deck slides	CO	Bloom
1	Finding patterns of text without regular expressions	Ch. 7	148-150	5, 6, 7	CO3	L3
2	Creating and matching Regex objects	Ch. 7	150-152	8, 9, 10	CO3	L3
3	Grouping with parentheses	Ch. 7	152-153	11, 12, 13	CO3	L3
4	Matching multiple groups with the pipe	Ch. 7	153-154	14, 15, 16	CO3	L4
5	Optional matching with the question mark	Ch. 7	154-155	17, 18, 19	CO3	L3
6	Matching zero or more with the star, one or more with the plus	Ch. 7	155-156	20, 21, 22	CO3	L3
7	Specific repetitions with curly brackets, and greedy versus nongreedy matching	Ch. 7	156-157	23, 24, 25	CO3	L4
8	The <code>findall()</code> method	Ch. 7	157-158	26, 27, 28	CO3	L4

#	Teaching unit	Chapter	Pages	Deck slides	CO	Bloom
9	Character classes and making your own	Ch. 7	158-159	29, 30, 31	CO3	L3
10	The caret, the dollar sign and the wildcard	Ch. 7	159-162	32, 33, 34	CO3	L4
11	Case-insensitive matching, sub() and managing complex regexes	Ch. 7	162-165	35, 36, 37	CO3	L4
12	Files and file paths: absolute, relative and the working directory	Ch. 8	173-177	38, 39, 40	CO3	L3
13	The os.path module: sizes, listings and validity	Ch. 8	177-180	41, 42, 43	CO3	L3
14	The file reading and writing process	Ch. 8	180-184	44, 45, 46	CO3	L4
15	Saving variables with the shelve module	Ch. 8	184-185	47, 48, 49	CO3	L4
16	Saving variables with pprint.pformat()	Ch. 8	185-192	50, 51, 52	CO3	L5

Verbatim syllabus lines

- Pattern Matching with Regular Expressions, Finding Patterns of Text Without Regular Expressions
- Finding Patterns of Text with Regular Expressions, More Pattern Matching with Regular Expressions
- The findall() Method, Character Classes, Making Your Own Character Classes
- The Caret and Dollar Sign Characters, The Wildcard Character, Review of Regex Symbols
- Reading and Writing Files, Files and File Paths, The os.path Module
- The File Reading/Writing Process, Saving Variables with the shelve Module
- Saving Variables with the pprint.pformat() Function
- Text book 1: Chapters 7, 8
- Number of Hours: 8

Provenance

Built from TB1 Chapters 7-8, which the syllabus names for this module. Chapter 7 is taught by first writing the phone-number matcher by hand so the cost of not having regular expressions is felt before the notation is introduced. The recurring theme in Chapter 8 is that a path is data too: os.path.join is preferred over string concatenation throughout, because the alternative breaks the moment the script moves between operating systems. shelve and pprint.pformat are presented as the two ways of persisting a Python structure without inventing a file format.

Module 4 — Classes and Objects

Driving question: When do I stop passing loose values around and define a type of my own — and what does a class give me that a dictionary does not?

Topic to chapter, pages and deck slides

#	Teaching unit	Chapter	Pages	Deck slides	CO	Bloom
1	Programmer-defined types	TB2	Ch. 15	5, 6, 7	CO4	L2
2	Attributes	TB2	Ch. 15	8, 9, 10	CO4	L3
3	Rectangles: objects as attributes of objects	TB2	Ch. 15	11, 12, 13	CO4	L4
4	Instances as return values	TB2	Ch. 15	14, 15, 16	CO4	L3
5	Objects are mutable	TB2	Ch. 15	17, 18, 19	CO4	L4
6	Copying objects	TB2	Ch. 15	20, 21, 22	CO4	L4
7	Classes and functions: the Time class	TB2	Ch. 16	23, 24, 25	CO4	L4
8	Pure functions	TB2	Ch. 16	26, 27, 28	CO4	L4
9	Modifiers	TB2	Ch. 16	29, 30, 31	CO4	L4
10	Prototyping versus planning	TB2	Ch. 16	32, 33, 34	CO4	L5
11	Classes and methods: object-oriented features	TB2	Ch. 17	35, 36, 37	CO4	L3
12	Printing objects and the <code>__init__</code> method	TB2	Ch. 17	38, 39, 40	CO4	L4
13	The <code>__str__</code> method	TB2	Ch. 17	41, 42, 43	CO4	L3
14	Operator overloading	TB2	Ch. 17	44, 45, 46	CO4	L4
15	Type-based dispatch and polymorphism	TB2	Ch. 17-18	47, 48, 49	CO4	L5
16	Choosing between a class, a dictionary and a tuple	TB2	Ch. 15-18	50, 51, 52	CO4	L5

Verbatim syllabus lines

- Classes and objects: Programmer-defined types, Attributes, Rectangles, Instances as return values, Objects are mutable, Copying
- Classes and functions: Time, Pure functions, Modifiers, Prototyping versus planning
- Classes and methods: Object-oriented features, Printing objects, Another example, The init method, The `__str__` method
- Operator overloading, Type-based dispatch, Polymorphism
- Text book 2: Chapters 15-18
- Number of Hours: 9

Provenance

The syllabus assigns this module entirely to TB2, Allen B. Downey, Think Python 2e, Chapters 15-18. That book was NOT supplied in this environment. Every unit below is therefore aligned to the verbatim syllabus headings, which are themselves Downey section titles, and cites only the chapter numbers the syllabus gives. No page numbers are claimed anywhere in this module, and every slide says so in its [Sources] note. Content is standard Python object semantics, which are verifiable against the language itself; a reviewer with the book should check section ordering before the module is taught.

Module 5 — Networked Programs, Web Services and Databases

Driving question: How does a Python program fetch data from somewhere else on the network, understand what it gets back, and store it so the next run does not have to fetch it again?

Topic to chapter, pages and deck slides

#	Teaching unit	Chapter	Pages	Deck slides	CO	Bloom
1	HTTP: the Hypertext Transfer Protocol	TB3	Ch. 12	5, 6, 7	CO5	L2
2	The world's simplest web browser	TB3	Ch. 12	8, 9, 10	CO5	L3
3	Retrieving an image over HTTP	TB3	Ch. 12	11, 12, 13	CO5	L4
4	Retrieving web pages with urllib	TB3	Ch. 12	14, 15, 16	CO5	L3
5	XML: eXtensible Markup Language	TB3	Ch. 13	17, 18, 19	CO5	L2
6	Parsing XML	TB3	Ch. 13	20, 21, 22	CO5	L4
7	Looping through nodes	TB3	Ch. 13	23, 24, 25	CO5	L3
8	JSON: JavaScript Object Notation	TB3	Ch. 13 up to 13.5	26, 27, 28	CO5	L3
9	What is a database, and why not a file?	TB3	Ch. 15	29, 30, 31	CO5	L3
10	Database concepts and the SQLite browser	TB3	Ch. 15	32, 33, 34	CO5	L3
11	Creating a database table	TB3	Ch. 15	35, 36, 37	CO5	L4
12	SQL: inserting, selecting, updating and deleting	TB3	Ch. 15	38, 39, 40	CO5	L4
13	Spidering a web API into a database	TB3	Ch. 15	41, 42, 43	CO5	L5
14	Basic data modelling and the three kinds of keys	TB3	Ch. 15	44, 45, 46	CO5	L5
15	Programming with multiple tables and JOIN	TB3	Ch. 15	47, 48, 49	CO5	L5
16	Putting it together: fetch, parse, store, query	TB3	Ch. 12-15	50, 51, 52	CO5	L5

Verbatim syllabus lines

- HTTP, The World's simplest Web Browser, Retrieving an image over HTTP, Retrieving web pages with urllib
- XML, Parsing XML, Looping through nodes, JSON
- What is database?, Database Concepts, Database Browser, Creating a database table
- SQL, Spidering Twitter, Basic data modeling
- Programming with multiple tables, Three kinds of Keys, JOIN
- Text book 3: Chapters 12, 13 (Upto 13.5), 15
- Number of Hours: 8

Provenance

The syllabus assigns this module entirely to TB3, Charles R. Severance, Python for Everybody, Chapters 12, 13 up to section 13.5, and 15. That book was NOT supplied in this environment. Every unit below is aligned to the verbatim syllabus headings, which are Severance section titles, and cites only the chapter numbers the syllabus gives; no page numbers are claimed and every slide says so in its [Sources] note.

Content is standard HTTP, XML/JSON and SQLite behaviour, verifiable against the Python standard library. The Twitter spidering example is presented as the book frames it, with a note that the public API it assumes has since been restricted.

Note on textbook page numbers

All TB1 chapter and page references were taken from the supplied Automate the Boring Stuff with Python PDF's own table of contents; the file extracted to roughly 1.8 million characters of machine-readable text, so no TB1 page reference in this document is from memory.

TB2 was not supplied in this environment. The syllabus lists Allen B. Downey, *Think Python: How to Think Like a Computer Scientist*, Green Tea Press, 2nd edition, 2015 for the whole of Module 4 (Classes and objects, Chapters 15-18). Units depending on it are aligned to the verbatim syllabus headings, cite the chapter numbers the syllabus itself gives, and claim no page numbers. Every affected slide says so in its [Sources] note.

TB3 was not supplied in this environment. The syllabus lists Charles R. Severance, *Python for Everybody: Exploring Data Using Python 3*, CreateSpace Independent Publishing Platform, 1st edition, 2016 for the whole of Module 5 (web services and databases, Chapters 12, 13 up to 13.5, and 15). Units depending on it are aligned to the verbatim syllabus headings, cite the chapter numbers the syllabus itself gives, and claim no page numbers. Every affected slide says so in its [Sources] note.